

Topic 7: Intermolecular Forces

Students will be assessed on their ability to:

7.1

understand the nature of the following intermolecular forces:

- London forces (instantaneous dipole-induced dipole)
- permanent dipole-permanent dipole interactions
- hydrogen bonds

London (Dispersion) Forces

This (**Instantaneous dipole-Induced dipole interaction**) occurs because in any molecule, the electrons are constantly and randomly moving around and so its electron density fluctuates over time which results in a temporary dipole being formed. These temporary dipoles induce more dipoles in neighbouring molecules. When this happens, the δ^+ end in one molecule and the δ^- end in the neighbouring molecule are attracted to each other

They occur **between all types of molecules** whether polar or non-polar and between separate atoms in noble gases. They do not occur in *ionic substances*

Main Factors that affect size of London Forces

- Number of electrons:** The more electrons there are in a molecule, the greater the fluctuations in electron density and so the larger the instantaneous and induced dipoles created. This makes London forces stronger between the molecules and more energy will be needed to break them apart so boiling point will increase
- Shape & Size of molecule:** The more points of contact there are between the molecules, the greater the overall London force as instantaneous dipole-induced dipole forces exist at each point of contact between the molecules. That is why Long straight chain alkanes have stronger London forces as they have a larger surface area of contact between molecules compared to spherical shaped branched alkanes

Permanent dipole-dipole forces

This (**Permanent dipole-Permanent dipole interaction**) is stronger than London forces and so their compounds have higher boiling points.

They occur between polar molecules.

The δ^+ and δ^- regions of neighbouring polar molecules attract each other and hold the molecules together in a lattice-like structure.

Polar bonds form due to a difference in electronegativity

Hydrogen Bonding

This is an intermolecular interaction (in which there is some evidence of bond formation) between a hydrogen atom of a molecule (or molecular fragment) bonded to an atom which is more electronegative than hydrogen and another atom in the same or a different molecule.

They act most often between compounds that have a hydrogen atom attached to one of the three most electronegative atoms, Nitrogen, Oxygen and Fluorine which all must have a lone pair of electrons

Hydrogen Bonding through Nitrogen

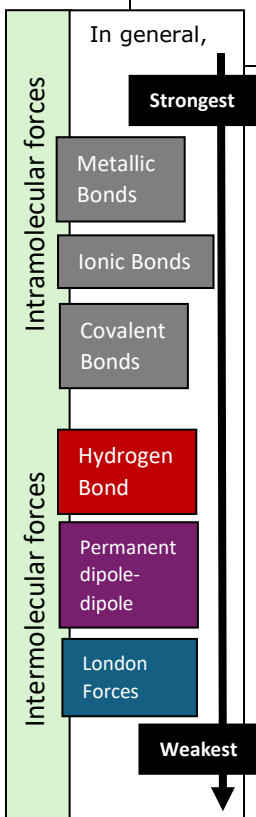
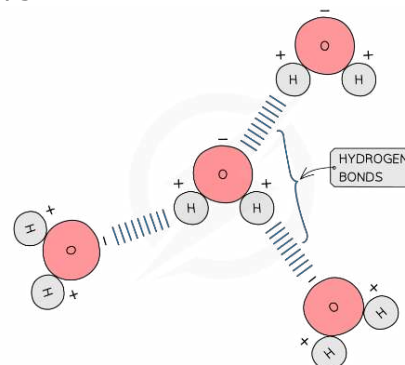
- All compounds containing an -N-H group can form intermolecular hydrogen bonds
- Primary Amines can form intermolecular hydrogen bonds

Hydrogen Bonding through Oxygen

- All compounds containing an -O-H group form intermolecular hydrogen bonds
- Alcohols form intermolecular hydrogen bonds

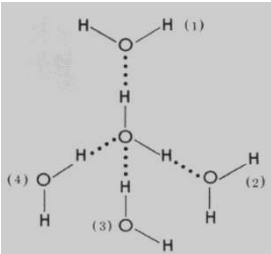
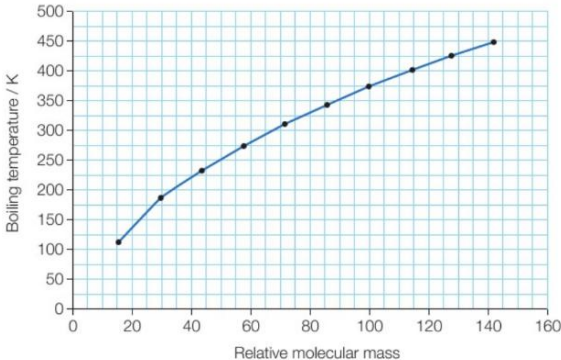
Hydrogen Bonding through Fluorine

- Only hydrogen fluoride forms intermolecular hydrogen bonds



It is better to assume that the strength of each type of intermolecular force depends on whether their interaction is favorable in each case

Note that **Hydrogen bonding** occurs in addition to **London forces**

7.2	understand the interactions in molecules, such as <i>*H₂O, liquid NH₃ and liquid HF*</i> , which give rise to hydrogen bonding																								
	Alcohols (O-H bond), ammonia (N-H bond), amines (N-H bond), carboxylic acids (O-H bond), hydrogen fluoride (H-F bond), proteins (N-H bond) can form hydrogen bonds																								
7.3	understand the following anomalous properties of water resulting from hydrogen bonding: - i its high melting and boiling temperature when compared with similar molecules - ii the density of ice compared to that of water																								
	- Water can form four hydrogen bonds per molecule, because the electronegative oxygen atom has two lone pairs of electrons on it and two hydrogen atoms. Thus, it can form stronger hydrogen bonds and needs more energy to break the bonds, leading to a higher melting and boiling point. - Density of ice at 0°C is less than that of water at 0°C because the molecules in ice are arranged in rings of six, held together by hydrogen bonds. The structure creates large areas of open space and when ice melts, the ring structure is destroyed and the average distance between molecules decreases, resulting in an increase in density. 																								
7.4	be able to predict the presence of hydrogen bonding in molecules analogous to those mentioned in 7.2																								
7.5	understand, in terms of intermolecular forces, physical properties shown by substances, including: - i the trends in boiling temperatures of alkanes with increasing chain length - ii the effect of branching in the carbon chain on the boiling temperatures of alkanes - iii the relatively low volatility (higher boiling temperatures) of alcohols compared to alkanes with a similar number of electrons - iv the trends in boiling temperatures of the hydrogen halides HF to HI																								
i	As chain length increases, - molecular mass increases and so does the number of electrons per molecule which causes an increase in instantaneous dipole-induced dipoles and - the number of contact points between molecules increases which means greater overall London forces, resulting in an increase in boiling point !Note there are only London forces in alkanes  <table border="1"> <caption>Approximate data points from the graph</caption> <thead> <tr> <th>Relative molecular mass</th> <th>Boiling temperature / K</th> </tr> </thead> <tbody> <tr><td>16</td><td>116</td></tr> <tr><td>30</td><td>182</td></tr> <tr><td>44</td><td>254</td></tr> <tr><td>58</td><td>271</td></tr> <tr><td>72</td><td>297</td></tr> <tr><td>86</td><td>309</td></tr> <tr><td>100</td><td>327</td></tr> <tr><td>114</td><td>340</td></tr> <tr><td>128</td><td>353</td></tr> <tr><td>142</td><td>369</td></tr> <tr><td>156</td><td>380</td></tr> </tbody> </table>	Relative molecular mass	Boiling temperature / K	16	116	30	182	44	254	58	271	72	297	86	309	100	327	114	340	128	353	142	369	156	380
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ii	The more branching in the molecule, the fewer the points of contact between adjacent molecules This leads to a decrease in overall intermolecular force of attraction between molecules and hence, a decrease in boiling temperature																								
iii	As alcohols contain an -OH group, they can form hydrogen bonds in addition to London forces. This means, compared to equivalent alkanes with similar chain length and number of electrons, the boiling point of alcohols would be higher due to the hydrogen bonding. This additional force of attraction increases the energy required to separate the molecules																								

	A direct measure of the strength of intermolecular interactions would be the enthalpy change of vaporization The greater the enthalpy change of vaporization, the greater the force of attraction between molecules	ΔH_{vap} is a measure of the energy change that is required to completely separate the molecules of a liquid and convert it into a gas at the same temperature																																										
iv	The anomalously high boiling points of H_2O, NH_3 and HF are due to the hydrogen bonding between the molecules in addition to the London forces already present, making it require more energy to break these forces of attraction The general increase in boiling point from HCl to HI is caused by the increasing number of electrons per molecule, thereby resulting in an increase in London forces All hydrogen halides have London forces and permanent dipole-permanent dipole forces between its molecules Only HF has hydrogen bonding in addition to these forces	<table border="1"> <caption>Approximate data from the Boiling point K vs Molecular mass graph</caption> <thead> <tr> <th>Compound</th> <th>Molecular mass</th> <th>Boiling point K</th> </tr> </thead> <tbody> <tr><td>CH_4</td><td>16</td><td>109</td></tr> <tr><td>SiH_4</td><td>32</td><td>161</td></tr> <tr><td>PH_3</td><td>34</td><td>186</td></tr> <tr><td>H_2S</td><td>34</td><td>213</td></tr> <tr><td>HCl</td><td>36.5</td><td>238</td></tr> <tr><td>H_2Se</td><td>81</td><td>241</td></tr> <tr><td>AsH_3</td><td>77.5</td><td>211</td></tr> <tr><td>HBr</td><td>81</td><td>207</td></tr> <tr><td>GeH_4</td><td>77</td><td>183</td></tr> <tr><td>SnH_4</td><td>151</td><td>221</td></tr> <tr><td>HI</td><td>128</td><td>238</td></tr> <tr><td>SbH_3</td><td>124</td><td>255</td></tr> <tr><td>H_2Te</td><td>129</td><td>271</td></tr> </tbody> </table>	Compound	Molecular mass	Boiling point K	CH_4	16	109	SiH_4	32	161	PH_3	34	186	H_2S	34	213	HCl	36.5	238	H_2Se	81	241	AsH_3	77.5	211	HBr	81	207	GeH_4	77	183	SnH_4	151	221	HI	128	238	SbH_3	124	255	H_2Te	129	271
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7.6	understand factors that influence the choice of solvents, including: - i water, to dissolve some ionic compounds, in terms of the hydration of the ions - ii water, to dissolve simple alcohols, in terms of hydrogen bonding - iii water, as a poor solvent for compounds (to include polar molecules such as halogenoalkane), in terms of inability to form hydrogen bonds - iv non-aqueous solvents, for compounds that have similar intermolecular forces to those in the solvent																																											
i	For a substance to dissolve, - The solute particles must be separated from each other and then become surrounded by solvent particles - The force of attraction between the solute and solvent particles must be strong enough to overcome the solvent-solvent forces and the solute-solute forces When an ionic lattice dissolves in water, the bonds in the lattice are broken and new bonds are formed between the ions and the water molecules The energy required to separate the ions in the solid is either completely or partially supplied by the hydration of ions The negative ions are attracted to the δ^+ hydrogen ends of the polar water molecules The positive ions are attracted to the δ^- oxygen ends of the polar water molecules In <i>general</i>, the greater the ionic charge, the less soluble an ionic compound is																																											
ii	The solubility of alcohols in water decreases with increasing hydrocarbon chain length as London forces predominate between the alcohol molecules																																											
iii	Compounds that cannot form hydrogen bonds with water include - Non-polar molecules such as the alkanes - Some polar molecules such as ethoxyethane and halogenalkanes																																											
iv	"Like dissolves like" meaning compounds that have similar intermolecular forces to those in the solvent will generally dissolve Non-polar solutes will dissolve in non-polar solvents (like hydrocarbons) Polar solutes generally dissolve in polar solvents																																											

	Further suggested practicals:
	i the solubility of simple molecules in different solvents
	ii measuring the enthalpy change of vaporisation of water
	iii measuring temperature changes when substances dissolve

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